

CE 310

Week 13 — Monte Carlo Simulation

Propagating Uncertainty Through a Model

University of Arizona · Dr. Wooyoung Jung · Fall 2026

From One Scenario to Many

Week 12 asked: "if outdoor temperature is 5°F too high, how does that change the prediction?" — one scenario, one answer.

The new question: inputs are not just biased by a fixed amount. They vary randomly from year to year. What is the full range of plausible annual outcomes?

Week 12 (OAT)

- One input, one shift
- Answer: a single Δ kWh number
- Cannot tell you probability
- Example: "+5°F → +45,100 kWh/yr"

Week 13 (Monte Carlo)

- All inputs simultaneously random
- Answer: a full distribution
- Gives probabilities and percentiles
- Example: "95% probability annual cost stays below \$24,000"

The Monte Carlo Method

Three steps:

1. **Model the input uncertainty:** assign each uncertain input a probability distribution (e.g., temperature anomaly $\sim \text{Normal}(0, 2^\circ\text{F})$)
2. **Sample:** draw N random values for each input ($N = 10,000$ is typical)
3. **Run the model for each sample:** produce N output values, forming an empirical distribution of outcomes

For a linear model this is fast — just array arithmetic. For complex building energy simulations (EnergyPlus, OpenStudio), each run may take minutes — Monte Carlo with $N = 10,000$ runs would require months.

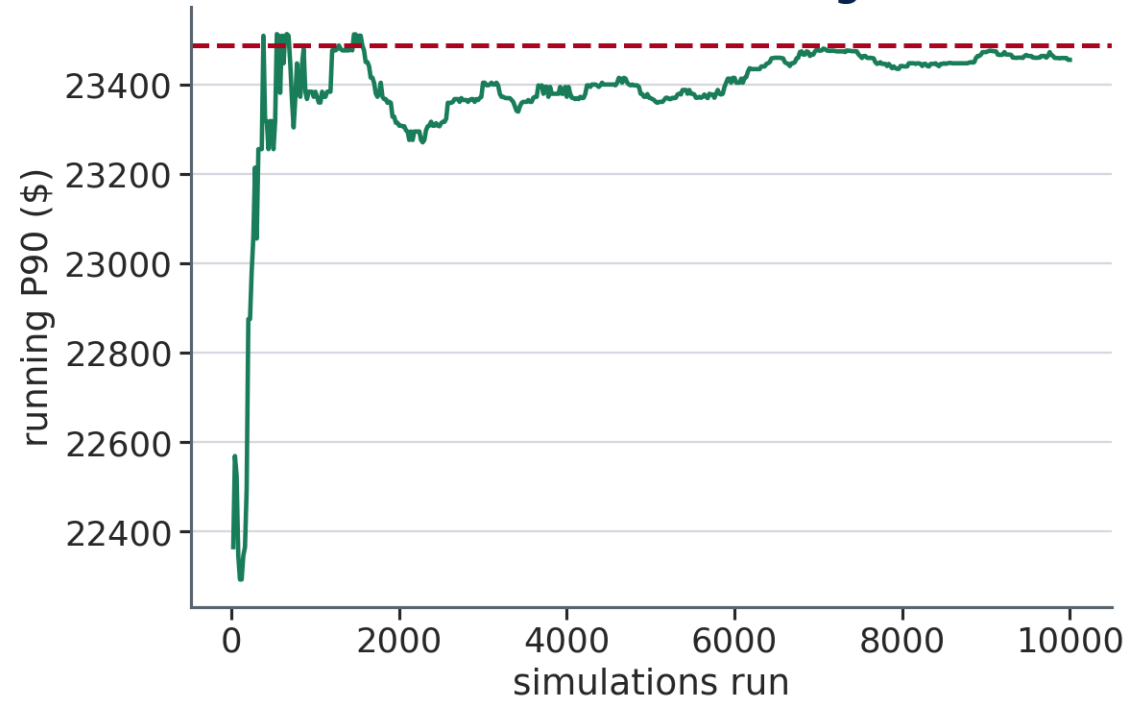
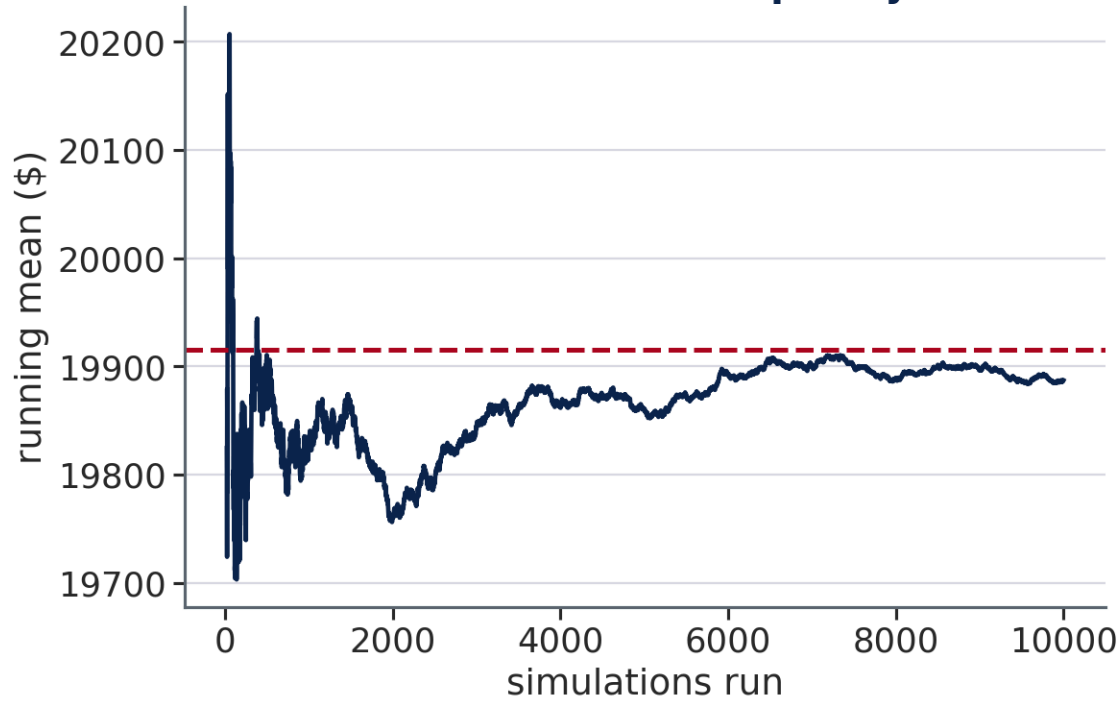
Why $N = 10,000$?

⚠ Why $N = 10,000$ and not $N = 100$? Tail percentiles (P5, P95) are estimated from only the most extreme samples. At $N = 100$, P5 is based on roughly 5 data points — noisy and highly seed-dependent. At $N = 10,000$, P5 is based on roughly 500 points — far more stable. The mean converges fast with small N ; the tails need many more samples to settle down.

Monte Carlo is not a special algorithm — it is just systematic sampling from uncertainty distributions. The "simulation" is the repeated evaluation of a model under random conditions.

Why 10,000 and Not 500

**Why $N = 10,000$ and not $N = 500$ — tails converge slowly
the mean settles quickly**



The mean stabilises after a few hundred runs. The P90 is still wandering at several thousand. Whatever you need from the tail, you need more runs to get it.

For Linear Models: Analytical vs. Simulated

For a linear model $Y = a + b \cdot X$ where $X \sim \text{Normal}(\mu, \sigma)$, the output std is:

$\sigma_Y = |b| \times \sigma_X$ — exact, no simulation needed.

Temperature example:

- Input: $\text{delta_T} \sim \text{Normal}(0, 2^\circ\text{F})$
- Analytical $\sigma_{\text{kWh}} = 2 \times 8,760 \times 1.026 = \mathbf{17,975 \text{ kWh/yr}}$
- Monte Carlo ($N = 10,000$) matches within a few hundred kWh

Monte Carlo is necessary when the model is non-linear or inputs are correlated — where the analytical formula breaks down.

The Output Distribution — What It Looks Like

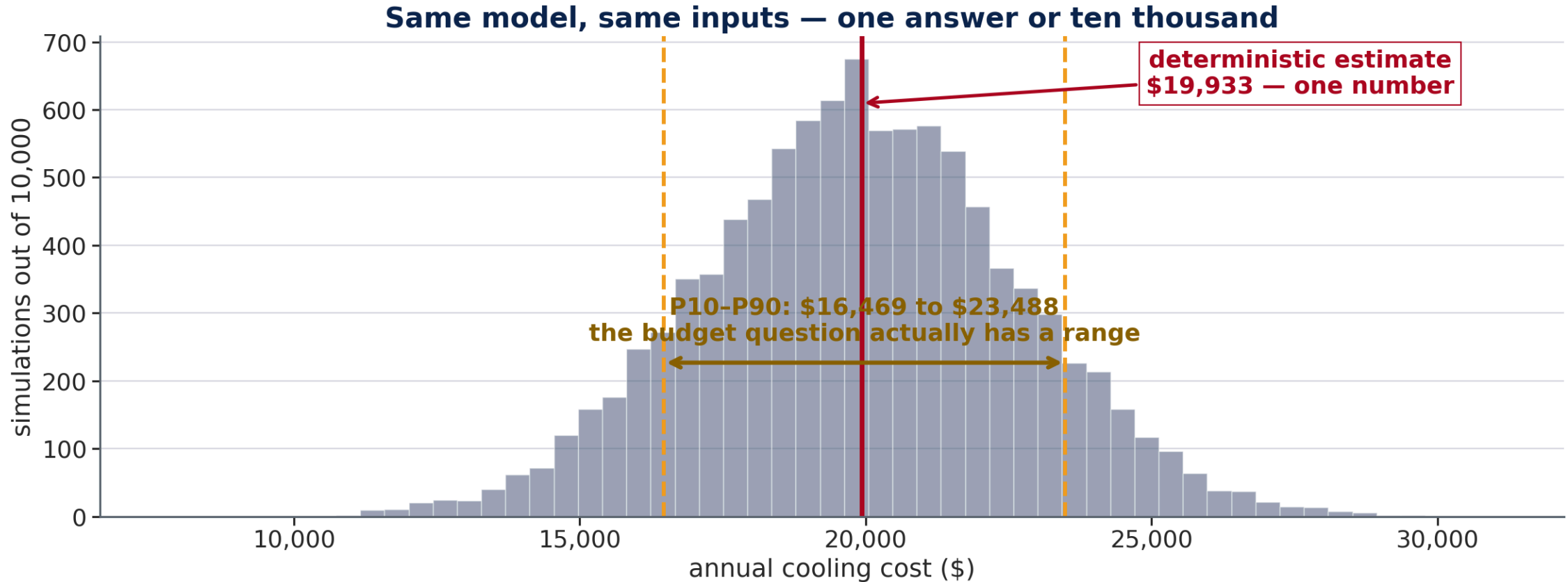
With three uncertain inputs (temperature, occupied hours, electricity rate), the annual cost distribution is approximately Normal by the Central Limit Theorem:

- **Mean** $\approx 166,107 \text{ kWh} \times \$0.12/\text{kWh} = \text{\$19,933/yr}$ (base case)
- **Std** $\approx \text{\$2,730/yr}$ (from all three inputs combined)
- **P5** $\approx \$15,440$ · **P95** $\approx \$24,425$

The histogram shows a bell curve centered near \$19,933. The tails represent low-probability but possible extreme years: an unusually hot summer (or high electricity price) could push costs toward the upper tail.

⚠ The output distribution is Normal here because the model is linear and all inputs are Normal. Non-linear models or non-Normal inputs can produce skewed, bimodal, or heavy-tailed output distributions that cannot be approximated analytically.

One Number or Ten Thousand



The deterministic estimate is not wrong. It is just the middle of a distribution the client never gets to see unless you compute it.

Exceedance Probabilities and Budget Decisions

Monte Carlo turns model output into decision-relevant probabilities.

Example: A facilities manager wants 95% confidence of staying on budget.

From the simulation:

- $P(\text{cost} > \$24,000) \approx \mathbf{6.8\%}$ — the baseline \$0.12/kWh rate is not safe
- **P95 $\approx$ \$24,425** — set the budget here for 95% confidence
- P5 $\approx$ \$15,440: below this in favorable years

Setting the budget at P95 means the reserve covers 95% of simulated years. You expect to spend less most years, but have coverage for rare expensive outcomes.

Quick Check — What Drives Cost Uncertainty?

60 seconds · rank the three inputs

Three inputs are uncertain in the cost model:

- **Outdoor temperature** — $\sigma = 2^{\circ}\text{F}$ around mean forecast
- **Occupied hours/year** — uncertain schedule ($\pm 5\%$)
- **Electricity rate** — $\pm 8\%$ around \$0.12/kWh

Last week temperature dominated **kWh** — but here the output is **annual energy cost**. **Which input contributes most to cost variance?**

- **A)** Temperature ($\sim 70\%$ of cost variance) — same winner as Week 12
- **B)** Electricity rate ($\sim 37\%$) — multiplying kWh by price amplifies its contribution
- **C)** Occupied hours ($\sim 50\%$)

✓ Temperature Dominates, But Rate Is Not Trivial

Multiplying by price changes the variance decomposition

Input	% of kWh variance (Week 12)	% of cost variance (Week 13)
Temperature	~100%	~62%
Electricity rate	0% (not in model)	~37%
Occupied hours	~0%	~1%

Why did rate jump from 0% to 37%? The regression predicts kWh — rate doesn't appear there. But annual cost = kWh × rate. An 8% rate swing on 166,000 kWh = ±13,280 kWh-equivalent of uncertainty — substantial.

💡 Uncertainty that matters depends on *what you're optimizing for* — a model accurate at predicting kWh can still produce highly uncertain cost forecasts when the price input is uncertain.

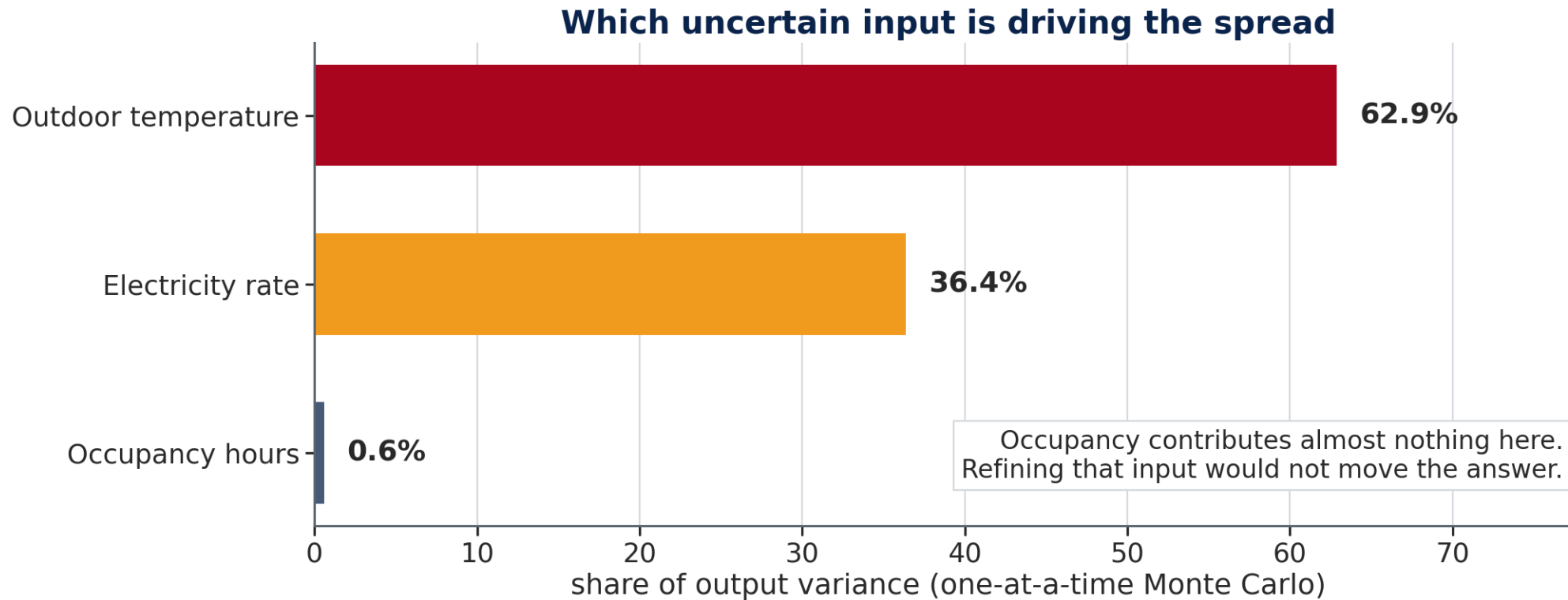
Variance Decomposition — What Drives the Uncertainty?

Run separate Monte Carlos where only one input varies at a time (other two held at mean). The variance from each run shows that input's contribution:

Input	% of Total Cost Variance
Outdoor temperature	~ 62%
Electricity rate	~ 37%
Occupied hours	~ 1%

Electricity rate contributes 37% of cost variance, but it was NOT in the Week 12 tornado diagram. The tornado diagram only perturbed regression model parameters — temperature slope, intercept, BH coefficient. Electricity rate is external to the regression model; it multiplies the output. Monte Carlo captures it; OAT sensitivity analysis did not.

Where the Spread Comes From



Temperature and rate together account for over 99% of the output variance. Occupancy contributes 0.62%, which is a finding worth acting on.

The Rate vs. Temperature Paradox

The variance decomposition shows electricity rate as the second-largest uncertainty driver — larger than occupancy by 60×. Yet rate has nothing to do with weather or building physics.

Why does rate matter so much? $\text{cost} = \text{kWh} \times \text{rate}$. Both kWh and rate are uncertain, and they multiply together. Even a $\pm 8\%$ rate variation ($\sigma_{\text{rate}} = 0.01$, $\text{mean} = 0.12$) amplifies through the full 166,107 kWh base.

The engineering lesson: sensitivity analysis on the regression model (Week 12) told you how cooling energy responds to weather and occupancy. Monte Carlo on the cost model adds the financial layer — and reveals that price risk can rival or exceed physical risk.

For a building energy analyst, this means: report both the **energy uncertainty** (which the regression model captures) and the **cost uncertainty** (which requires adding a price model).

Quick Check — Will the Budget Hold?

90 seconds · estimate the risk

Monte Carlo results:

- Baseline cost: **\$19,933/yr**
- Standard deviation: **\$2,730/yr** (approximately Normal)

A facilities manager's budget cap is **\$24,000/yr**.

Estimate: what is the probability of exceeding the \$24,000 budget? (*Hint: how many σ above the mean is \$24,000?*)

- **A)** $< 5\%$ — the budget has comfortable margin
- **B)** $\approx 6\text{--}8\%$ — marginal risk, but real
- **C)** $> 15\%$ — high risk, budget needs revision

✓ Answer: B — About 6.8% Exceedance Risk

The budget is closer to the edge than it looks

$$z = \frac{24,000 - 19,933}{2,730} = 1.49 \quad P(Z > 1.49) \approx \mathbf{6.8\%}$$

Budget	z-score	Exceedance probability
\$22,000	0.76	22.4% — high risk
\$24,000	1.49	6.8% — marginal
\$24,425 (P95)	1.65	5.0% — just at threshold
\$25,000	1.86	3.1% — comfortable

Decision: To guarantee < 5% exceedance risk, the budget must be at least **\$24,425**.

What Budget Level Keeps Risk Acceptable?

Monte Carlo directly answers: "what electricity rate would keep $P(\text{cost} > \$24,000)$ below 5%?"

By searching for the rate threshold:

- At \$0.12/kWh: $P(\text{cost} > \$24\text{k}) \approx 6.8\%$ — above target
- Threshold $\approx$ **\$0.116/kWh** ($\approx 3.3\%$ lower): $P(\text{cost} > \$24\text{k})$ drops to $< 5\%$

This is an actionable result: if the energy manager can negotiate a fixed-rate contract at $\leq$ \$0.116/kWh, the 5% budget exceedance target is met. Monte Carlo converts uncertainty into a procurement decision.

Bridge to Machine Learning

Monte Carlo concept	ML equivalent	Connection
Input distribution	Feature distribution	How the input varies in real data
N = 10,000 draws	Bootstrapping	Repeated sampling from a data source
Output distribution	Prediction interval	Full uncertainty of a model's outputs
Variance decomposition	Feature importance (SHAP)	Which input drives output variance?
P95 threshold	Safety margin	Risk-calibrated design specification

Looking Ahead — Week 14: Bootstrap Confidence Intervals

Monte Carlo simulates uncertainty by sampling from **assumed distributions** (Normal, etc.). But what if you don't know the distribution?

Bootstrap resamples from the **observed data** itself — no distributional assumption needed.

To estimate the uncertainty in a regression slope:

1. Draw a sample of rows (with replacement) from the dataset
2. Fit the regression on this resample
3. Record the slope
4. Repeat 10,000 times → distribution of slope values → confidence interval

Bootstrap applies to any statistic: slopes, percentiles, correlation coefficients, R^2 values.